

The modified diagonal cycles of hypergeometric curves

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§1 Notation

X : smooth projective var. / $\mathbb{C}$

$CH_k(X)$: Chow group of dim k on $X := \{ \text{algebraic cycles} \} / \sim_{\text{rat.}}$

$$= \{ \sum n_i Y \mid n_i \in \mathbb{Z}, Y \subseteq X : \text{irr closed dim } Y = k \}$$

$$cl : CH_k(X) \longrightarrow H_{2k}(X, \mathbb{Z}) : \text{cycle map}$$

U)

$$\ker(cl) =: CH_k(X)_{\text{hom}}$$

$$(e.x. \ X: \text{curve} \ CH_0(X)_{\text{hom}} = \text{Pic}(X)^0 \simeq \text{Jac}(X))$$

• generalized BSD

$$\text{rank } CH_k(X)_{\text{hom}} \stackrel{?}{=} \text{ord}_{s=k+1} (h^{2k+1}(X), s)$$

Hodge filtration

$$\left(\begin{array}{l} X: \text{curve} \\ \Phi_0 : CH_0(X)_{\text{hom}} \xrightarrow{\sim} J_0 \end{array} \right)$$

Φ_k : Abel-Jacobi map

$$\Phi_k : CH_k(X)_{\text{hom}} \longrightarrow J_k(X) := \frac{(\mathbb{F}^{k+1} H^{2k+1}(X, \mathbb{C}))^\vee}{H_{2k+1}(X, \mathbb{Z})}$$

C : smooth projective curve / $\mathbb{C}$

$e \in C$: fix.

$\rightsquigarrow$ Construct $CH_1(C^3)_{\text{hom}} !!$

Definition

$$\Delta_{\text{GKS},e}(C) := \Delta_{123} - \Delta_{12} - \Delta_{23} - \Delta_{13} + \Delta_1 + \Delta_2 + \Delta_3.$$

$$\Delta_{123} := \{ (x_1, x_2, x_3) \mid x_1 = x_2 = x_3 \in C \}$$

$$\Delta_{12} := \{ (x, x, e) \mid x \in C \}, \quad \Delta_{23}, \Delta_{13} \text{ も同様.}$$

$$\Delta_1 := \{ (x, e, e) \mid x \in C \}. \quad \Delta_2, \Delta_3 \text{ も同様}$$

$$\leadsto \Delta_{\text{GKS},e}(C) \in \text{CH}_1(C^3)_{\text{hom}}$$

§2 known results

(i) (Gross - Schoen '95)

$C: g=1, 2$, $\Delta_{\text{GKS},e}(C)$ is torsion

(ii) (Ceresa '83, Colombo - van Geemen '93)

For a generic $C (g \geq 3)$, $\Phi_1(\Delta_{\text{GKS},e}(C))$ is non-torsion

(iii) (Bloch '84)

$C = F_4$ (Fermat 4次)

(iv) (Kimura '00)

$C = \text{Klein curve}$

(v) (Eskandari - Murty '21)

$C = F_N$ ($\exists p > 7$: prime, such that $p \mid N$)

(vi) (N' 24)

$C = F_N$ の商

C = modular curve

§3 Main result

$$S = \mathbb{P}^1 \setminus \{0, \infty, 1\} \quad N \geq 2 \in \mathbb{Z}.$$

$$f_N : X_N \longrightarrow S \underset{\lambda}{\cup} : \text{projective curve}$$

$$f_N^{-1}(\lambda) : (x_2^N - x_1^N)(y_2^N - y_1^N) = \lambda x_1^N y_1^N \quad \text{in } \mathbb{P}^1 \times \mathbb{P}^1.$$

$$\text{affine model} \rightsquigarrow x := \frac{x_1}{x_2}, \quad y := \frac{y_1}{y_2} \quad \leftarrow \begin{array}{l} \text{超幾何曲線} \\ (\text{Asakura - Otsuka}) \end{array}$$

$$(1 - x^N)(1 - y^N) = \lambda x^N y^N$$

- $X_{N,\lambda} := f_N^{-1}(\lambda) : \text{non-sing. genus} = (N-1)^2$
- periods $\iff {}_2F_1\left(\frac{N}{2}, \frac{b}{N} : \lambda\right), \quad 1 \leq a, b < N.$
- $\# X_{N,\lambda}(\mathbb{F}_q) \iff {}_2F_1(\text{---})_{\mathbb{F}_q}.$
- $X_{N,\lambda}/\mathbb{Q}_p \iff \text{Frobenius 固有値 } {}_2F_1 \text{ p-adic}.$

Theorem 1 [Eskandari - N.]p23 : prime $e \in X_{N,\lambda} : \text{cusp}$

Then, $\forall \lambda \in \mathbb{P}^1 \setminus \{0, 1, \infty\}$ and $\forall \lambda \leq \frac{p-1}{2}, \quad \ell \Phi_1(\Delta_{\text{GKS}, e}(X_{p,\lambda})) \neq 0$

Theorem 2

$\forall \lambda \in S, \forall e \in X_{3,\lambda} \cdot \Delta_{GKS,e}(X_{3,\lambda})$ is torsion.

Sketch of Proof

$$\begin{array}{ccc} \Delta_{GKS} \in CH_1(X^3)_{hom} & \xrightarrow{\pi_Z} & CH_0(X)_{hom} \\ \downarrow \overline{\Phi}_1 & \wr & \downarrow \overline{\Phi}_0 \\ J_1(X) & \xrightarrow{\quad} & \underline{J_0(X) = J_{ac}(X)} \end{array}$$

Jacobi に 帰着 させろ

$$\begin{array}{ccc} & X^3 \times X & \\ \swarrow Pr_{123} & & \searrow Pr_4 \\ X^3 & & X \end{array} \quad \begin{array}{l} \Delta(X) \in CH_1(X \times X) \\ \sim \exists P \in CH_1(X \times X) \\ Z := \Delta(X) \times P \in CH_2(X^3 \times X^3) \end{array}$$

$$\begin{array}{ccc} \pi_Z : CH_1(X^3)_{hom} & \longrightarrow & CH_0(X)_{hom} \\ \wr & \longmapsto & Pr_4^* (Pr_{123}^* \Omega \cdot Z) \end{array}$$

$$P : d_0 \eta^{\sim 77}$$

$$\begin{array}{ccc} d : X & \longrightarrow & X \\ (x,y) \longmapsto & \left(\frac{1}{p_x}, \frac{1}{p_y} \right), & p = (1-\lambda)^{\frac{1}{N}} \end{array}$$